

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 23: Probability

Every formula and result of the chapter on one sheet

|           |                                 |                                                                   |
|-----------|---------------------------------|-------------------------------------------------------------------|
| <b>1</b>  | <b>Probability</b>              | $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ |
| <b>2</b>  | <b>Range</b>                    | $0 \leq P(E) \leq 1$                                              |
| <b>3</b>  | <b>Impossible event</b>         | $P = 0$                                                           |
| <b>4</b>  | <b>Sure event</b>               | $P = 1$                                                           |
| <b>5</b>  | <b>Complement</b>               | $P(\text{not } E) = 1 - P(E)$                                     |
| <b>6</b>  | <b>One coin</b>                 | 2 outcomes                                                        |
| <b>7</b>  | <b>Two coins</b>                | 4 outcomes                                                        |
| <b>8</b>  | <b>Three coins</b>              | 8 outcomes                                                        |
| <b>9</b>  | <b>One die</b>                  | 6 outcomes                                                        |
| <b>10</b> | <b>Two dice</b>                 | 36 outcomes                                                       |
| <b>11</b> | <b>Pack of cards</b>            | 52 cards, 4 suits of 13, 12 face cards                            |
| <b>12</b> | <b>Experimental probability</b> | $\frac{\text{times the event happened}}{\text{number of trials}}$ |